Chapter 15 (Oscillations) — In-Class Practice

Practice for HW 15.19 — spring constant / compression from a collision with a spring

Simple. (similar to 15.18) A 2 kg block is attached to a spring with a spring constant of 20 N/m. While the block is sitting at rest, a student hits it with a hammer and it almost instantaneously speeds up to 60 cm/s.
a. What is the amplitude of the subsequent oscillations?
b. What is the speed of the block at the point where $x = \frac{1}{2}A$?

Challenge. (similar to 15.63) A 500 g air-track glider attached to a spring with spring constant 10 N/m is sitting at rest on a frictionless air track. A 250 g glider is pushed toward it from the far end of the track at a speed of 120 cm/s. It collides with and sticks to the 500 g glider. What are the amplitude and period of the subsequent oscillations?

Practice for HW 15.23 — finding mass/spring constant from an oscillation count

Simple. (similar to 15.14) A 200 g air-track glider is attached to a spring. The glider is pushed in 10 cm and released. A student with a stopwatch finds that 10 oscillations take 12.0 s. What is the spring constant?

Challenge. (similar to 15.21) A spring is hanging from the ceiling. Attaching a 500 g physics book to the spring causes it to stretch 20 cm in order to come to equilibrium.

a. What is the spring constant?

b. From equilibrium, the book is pulled down 10 cm and released. What is the period of oscillation?

c. What is the book's maximum speed?

Practice for HW 15.24 — simple pendulum period relationships

Simple. (similar to 15.25) A pendulum is made by tying a 75 g ball to a 130-cm-long string. The ball is pulled 5.0° to the side and released. How many times does the ball pass through the lowest point of its arc in 7.5 s?

Challenge. (similar to 15.31) A uniform steel bar swings from a pivot at one end with a period of 1.2 s. How long is the bar? (This is a physical pendulum — use the moment of inertia of a rod pivoted at one end.)

Practice for HW 15.28 — relating a pendulum's swing angle and its speed (energy conservation)

Simple. (similar to 15.59) Orangutans can move by brachiation, swinging like a pendulum beneath successive handholds. If an orangutan has arms that are 0.90 m long and repeatedly swings to a 20° angle, taking one swing after another, estimate its speed of forward motion in m/s. While this is somewhat beyond the range of validity of the small-angle approximation, the standard results for a pendulum are adequate for making an estimate.

Challenge. (written for this class) A 20 g dart is thrown horizontally and embeds itself in a 480 g wood block that hangs at rest from a 90-cm-long string, so that the dart and block together swing as a pendulum. If the pendulum swings up to a maximum angle of 15° from vertical, how fast was the dart moving just before it hit the block?

Practice for HW 15.49 — full SHM analysis from initial position/velocity

Simple. (similar to 15.20) A ball oscillating on a spring with spring constant 45 N/m has a maximum displacement of 18 cm. The ball's speed is 3.3 m/s when its displacement is 12 cm. What is the ball's mass in g?

Challenge. (similar to 15.44) A 500 g mass on a horizontal spring oscillates with a period of 2.0 s and an amplitude of 75 cm. During one oscillation, at the instant the spring is at maximum extension, the mass explodes into two pieces. One piece is shot forward, away from the spring, at 2.0 m/s. The second piece, which remains attached to the spring, recoils at 3.0 m/s. After the explosion, what are the (a) period and (b) amplitude of the oscillation?